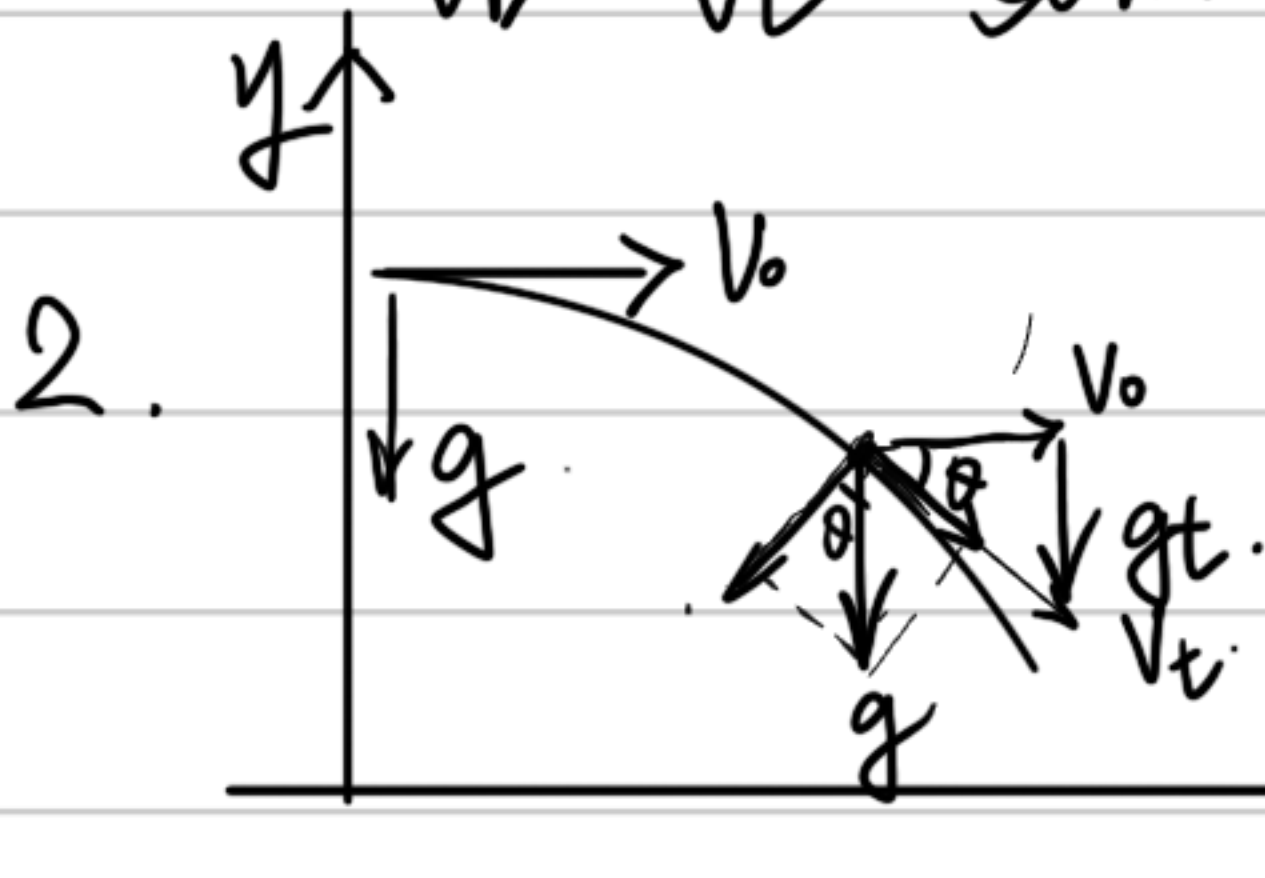


Problem Set 2

1. (a) Since the nod on top of B is part of the rope with $V = 100 \text{ mm/s}$ so $V_B = 100 \text{ mm/s}$ with $\vec{V}_B$ direction upward.

(b) And since the rope lifting up L and the pulley above L is related to B, the speed of the end of the rope is twice the speed of the pulley, so $V_L = \frac{1}{2} V_B = 50 \text{ mm/s}$ with $\vec{V}_L$ direction upward.
 $V_B - V_L = 50 \text{ mm/s}$ with the direction upward.



in the Cartesian coordinate system;

we obtain 2 right-angle triangle;

at the instant t , the angle between V_t and

the horizontal line is θ

so we get 2 similar right-angle triangles

where $\tan \theta = \frac{gt}{V_0}$ $\theta = \arctan \frac{gt}{V_0}$

And since $\vec{g}$ and $\vec{V}_0$ are perpendicular; $\alpha_z \perp \alpha_n$

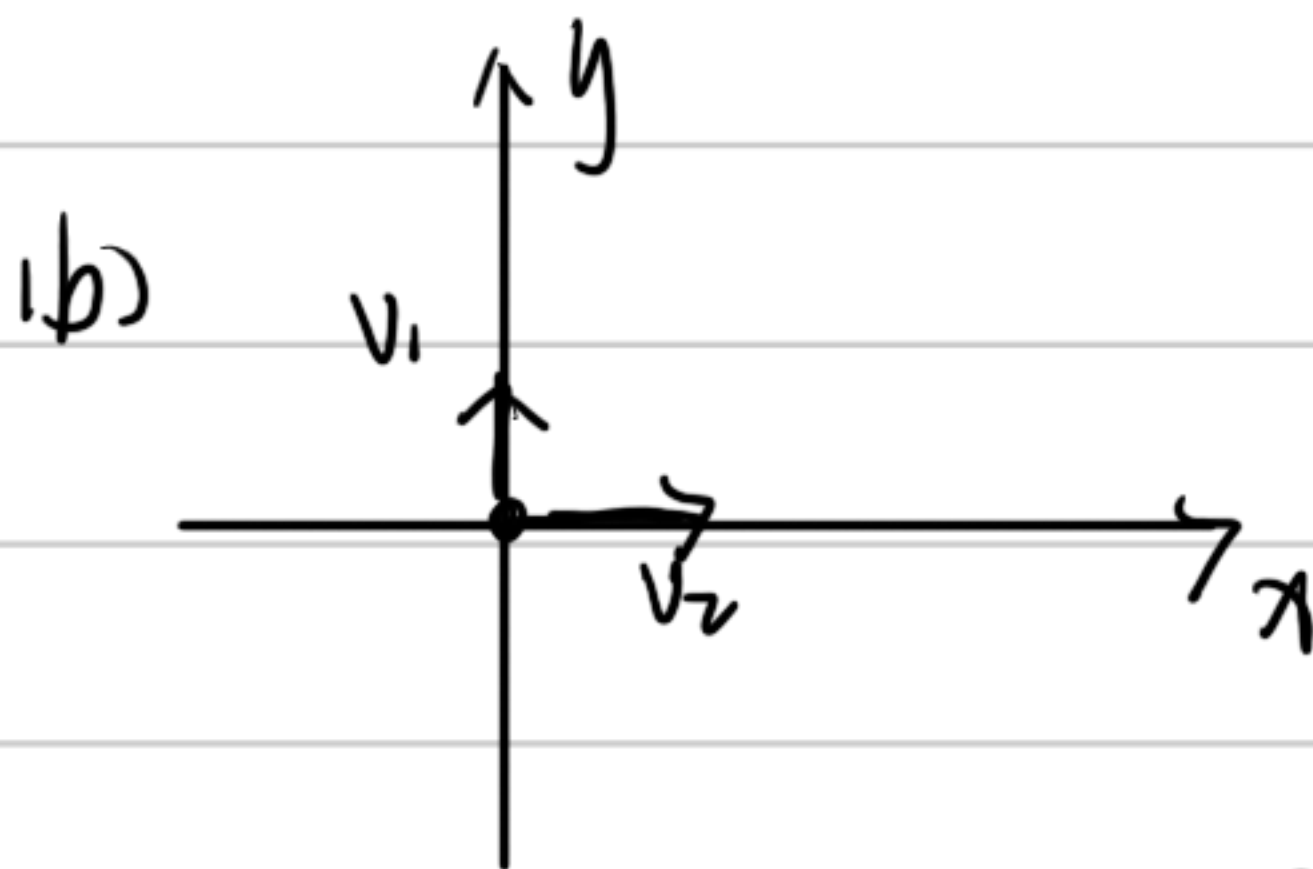
$\alpha_z = g \sin \theta$ $\alpha_n = g \cos \theta$ since $\theta = \arctan \frac{gt}{V_0}$

so $\alpha_z = \frac{g^2 t}{V_0^2 + g^2 t^2}$ $\alpha_n = \frac{g V_0}{\sqrt{V_0^2 + g^2 t^2}}$
 with direction = $\angle(\vec{\alpha}_z, x\text{-axis}) = \theta$ $\angle(\vec{\alpha}_n, -y\text{-axis}) = \theta$ $\theta = \arctan \frac{gt}{V_0}$

3. $V_1 \uparrow$ $V_2(y)$

(a) $|V| = \sqrt{V_1^2 + V_2^2} = \sqrt{V_1^2 + V_0^2 \sin^2 \frac{\pi y}{L}}$

$\vec{V}$ direction: the angle between $\vec{V}$ and bank is $\arctan \frac{V_1}{V_0 \sin \frac{\pi y}{L}}$



Set the original point to be $(0, 0)$

$x = \int_0^t V_2 dt = \int_0^t V_0 \sin \frac{\pi y}{L} dt$

$y = V_1 t$

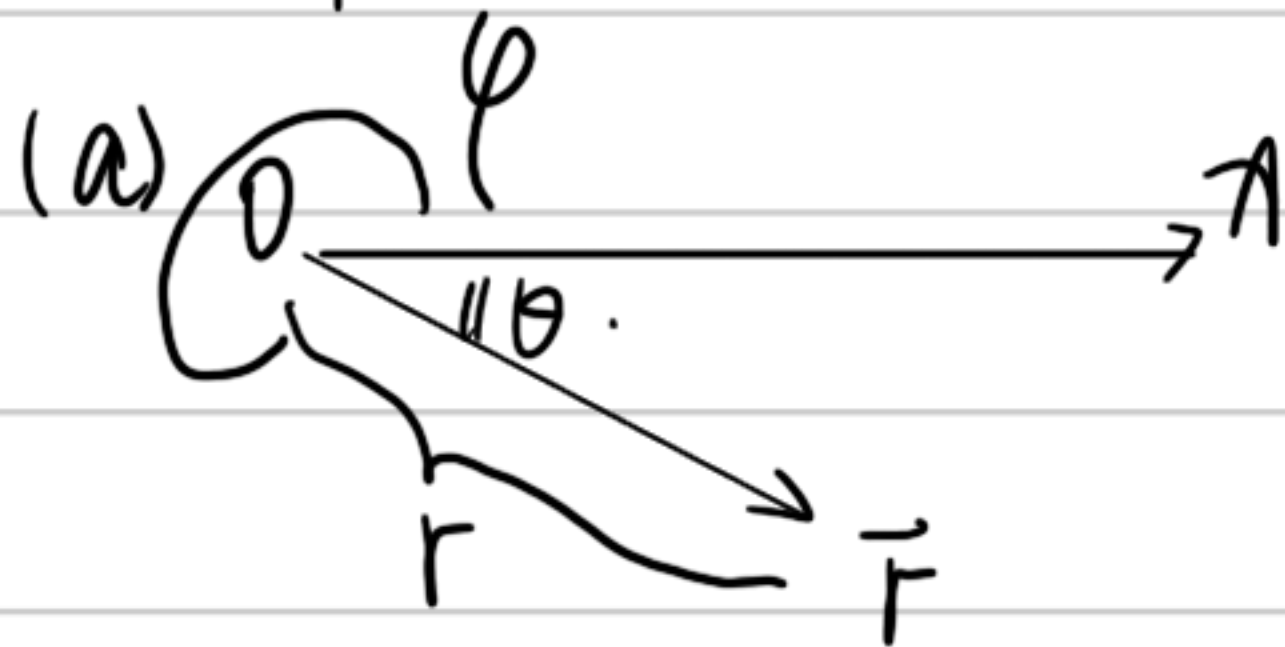
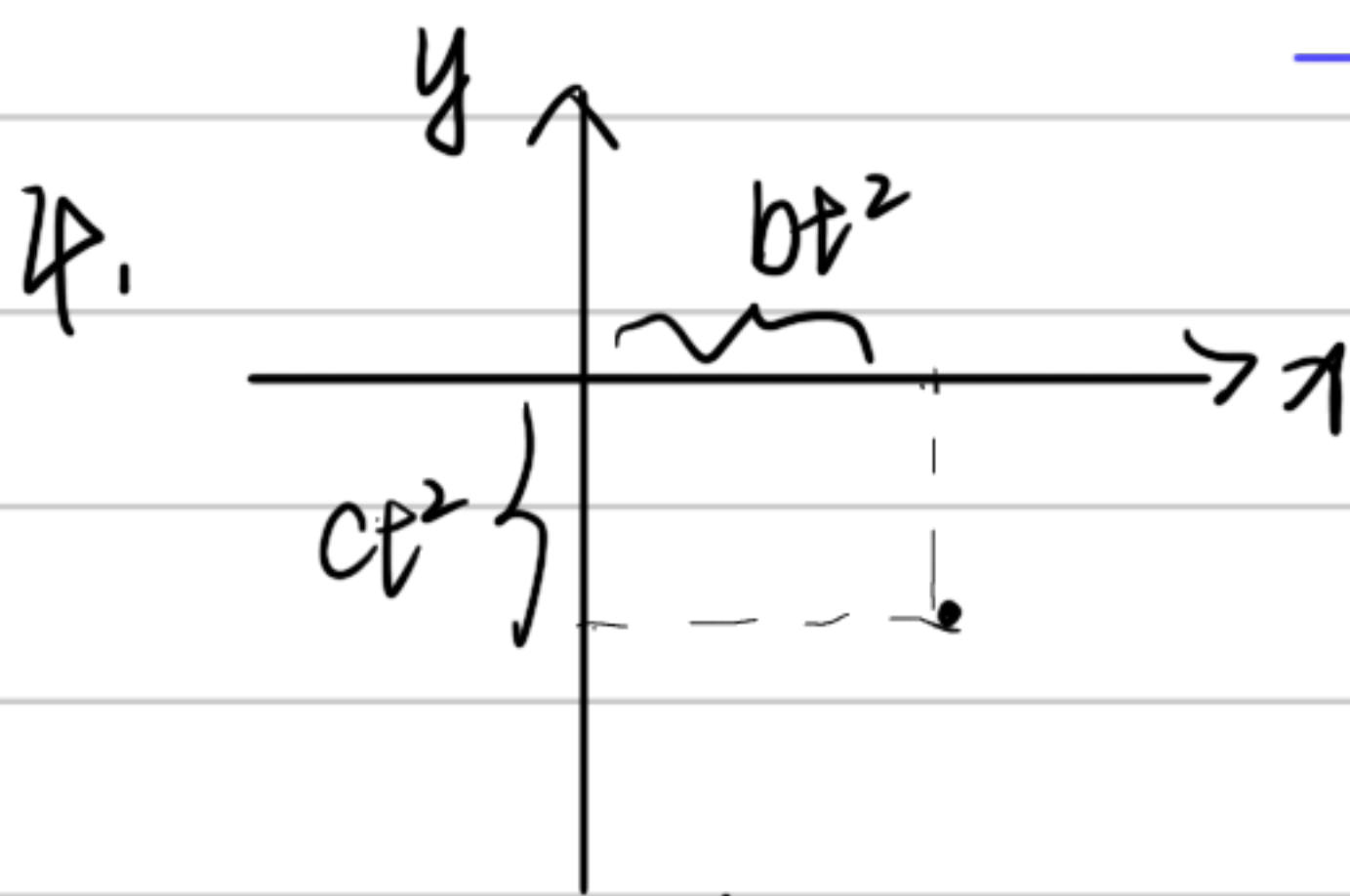
$x = \int_0^t V_0 \sin \frac{\pi y}{L} dt = \int_0^t V_0 \sin \left(\frac{\pi V_1}{L} t \right) dt =$

$= -\cos \left(\frac{\pi V_1}{L} t \right) \frac{L V_0}{\pi V_1} + \frac{L V_0}{\pi V_1}$ $y = \frac{L}{\pi} \arccos \left(-\frac{\pi V_1 x}{L V_0} \right)$

so trajectory: $\begin{cases} x = -\cos \left(\frac{\pi V_1}{L} t \right) \frac{L V_0}{\pi V_1} + \frac{L V_0}{\pi V_1} \\ y = V_1 t \end{cases}$

$x = -\cos \left(\frac{\pi V_1}{L} \frac{y}{V_1} \right) \frac{L V_0}{\pi V_1} + \frac{L V_0}{\pi V_1}$

(c) to cross the river: $y = V_1 t = L$ $t = \frac{L}{V_1}$
 so $x = -\cos\left(\frac{\pi V_1}{L} \cdot \frac{L}{V_1}\right) \frac{LV_0}{\pi V_1} + \frac{LV_0}{\pi V_1} = \frac{2LV_0}{\pi V_1}$
 the distance $d = \sqrt{x^2 + y^2} = \sqrt{\frac{4L^2 V_0^2}{\pi^2 V_1^2} + L^2}$



$$r = \sqrt{(bt^2)^2 + (ct^2)^2} = \sqrt{b^2 t^4 + c^2 t^4} = t^2 \sqrt{b^2 + c^2}$$

$$\varphi = 2\pi - \arctan \frac{c}{b}$$

so $r = \sqrt{b^2 t^4 + c^2 t^4}$
 $\varphi = 2\pi - \arctan \frac{c}{b}$

(b) in polar coordinate system:

since the direction of the particle keeps the same

implicit equation: $\varphi = 2\pi - \arctan \frac{c}{b}$ $r \geq 0$

in Cartesian system:

since $\begin{cases} x = bt^2 \\ y = -ct^2 \end{cases}$ $\frac{y}{x} = -\frac{c}{b}$

so $y = -\frac{c}{b}x$ $x \geq 0$

(c) in Cartesian system:

$V_x = \dot{x}(t) = 2bt$ $V_y = \dot{y}(t) = -2ct$ $v = \sqrt{V_x^2 + V_y^2} = 2t\sqrt{b^2 + c^2}$

with direction $\arctan \frac{c}{b}$ deviated from +x axis towards -y axis

$a_x = \dot{V}_x = 2b$ $a_y = \dot{V}_y = -2c$ $a = \sqrt{a_x^2 + a_y^2} = 2\sqrt{b^2 + c^2}$

with direction $\arctan \frac{c}{b}$ deviated from +x axis towards -y axis

in polar system: since the direction doesn't change

$v = \dot{r} = 2t\sqrt{b^2 + c^2} \hat{r}$ both direction φ counter clockwise

$a = \dot{v} = 2\sqrt{b^2 + c^2} \hat{r}$ deviated from x axis $\varphi = 2\pi - \arctan \frac{c}{b}$

5. (a) $r(t) = r_0(1 - ct) \Rightarrow r_0 - r(t) = r_0 ct \quad t = \frac{r_0 - r(t)}{r_0 c}$

so $\varphi(t) = \frac{1 - \frac{r(t)}{r_0}}{\frac{r(t)}{r_0}} = \frac{r_0 - r(t)}{r(t)}$

(b) $\vec{v} = \dot{\vec{r}} = \dot{r}(t) \hat{n}_r + r \dot{\varphi}(t) \hat{n}_\varphi$
 $= -r_0 c \hat{n}_r + \frac{c}{(1-ct)^2} r_0(1-ct) \hat{n}_\varphi = -r_0 c \hat{n}_r + \frac{c r_0}{1-ct} \hat{n}_\varphi$

radial component: $-r_0 c \hat{n}_r$

transverse component: $\frac{c r_0}{1-ct} \hat{n}_\varphi$

speed $v = |\vec{v}| = \sqrt{r_0^2 c^2 + \left(\frac{c r_0}{1-ct}\right)^2}$

$= r_0 c \sqrt{1 + \frac{1}{(1-ct)^2}}$

(c) $\vec{a} = \ddot{\vec{r}} = \ddot{r}(t) \hat{n}_r + \dot{r}(t) \dot{\hat{n}}_r + \dot{r}(t) \dot{\varphi}(t) \hat{n}_\varphi + r \ddot{\varphi}(t) \hat{n}_\varphi + r \dot{\varphi}(t) \dot{\hat{n}}_\varphi$
 $\dot{\hat{n}}_r = \dot{\varphi} \hat{n}_\varphi \quad \dot{\hat{n}}_\varphi = -\dot{\varphi} \hat{n}_r$

$\vec{a} = \ddot{\vec{r}} = -r_0 c \dot{\varphi} \hat{n}_\varphi + \frac{\dot{c} r_0}{(1-ct)^2} \hat{n}_\varphi - \dot{\varphi} \frac{c r_0}{1-ct} \hat{n}_r$
 $= -r_0 c \frac{c}{(1-ct)^2} \hat{n}_\varphi + \frac{c^2 r_0}{(1-ct)^2} \hat{n}_\varphi - \frac{c^2 r_0}{(1-ct)^3} \hat{n}_r$

$= 0 \hat{n}_\varphi - \frac{c^2 r_0}{(1-ct)^3} \hat{n}_r$

$+ \ddot{r}_0 (1-ct)^{-2}$

radial component: 0

transverse component: $\frac{c r_0 - c^2 r_0}{(1-ct)^2} \hat{n}_\varphi$

(d) when $c > 0$

$r(t)$ will decrease with t increasing;

$\varphi(t)$ will increase with t increasing.

the particle moves in a counterclock direction, getting closer to (0,0) $0 \leq t < \frac{1}{c}$

when $c < 0$

$r(t)$ will increase with t increasing;

$\varphi(t)$ will decrease with t increasing.

the particle moves in a clockwise direction, getting away from (0,0)

6. (a) unit for ω inside $\sin(\omega t)$ = s^{-1}

$$2 : s^{-1}$$

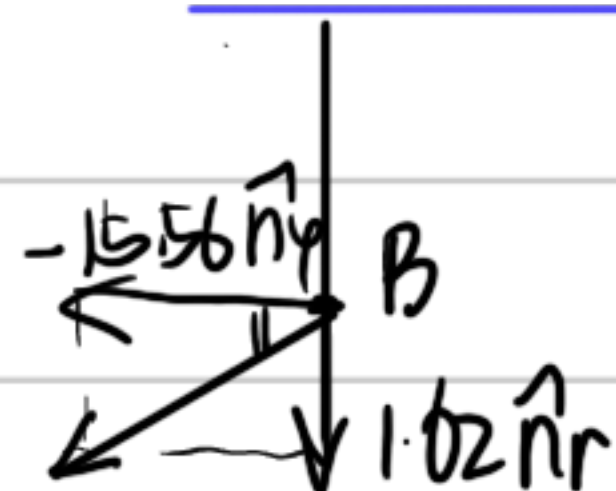
$$b : \text{cm.}$$

1. e. 3 = no unit

(b) $\vec{v} = \dot{\vec{r}} = \dot{r}\hat{n}_r + r\dot{\varphi}\hat{n}_\varphi = 12e^{-2t}\hat{n}_r + 6(1-e^{-2t})3\cos(\pi t)\hat{n}_\varphi$

$t=1s$ $\vec{v} = 1.62\hat{n}_r - 15.56\hat{n}_\varphi \text{ cm/s}$

direction



the angle between $\vec{v}$ and horizontal line is

$\arctan 0.104$ downwards

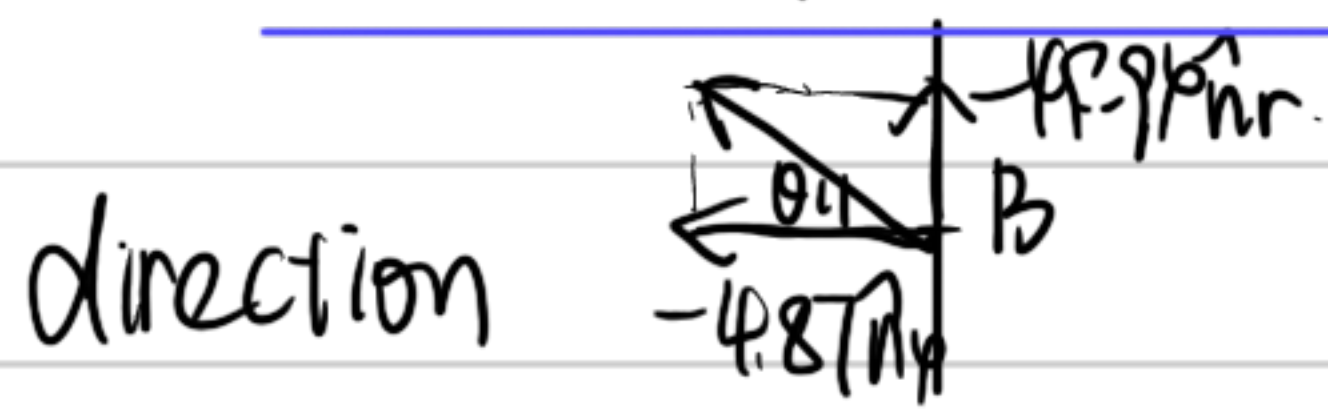
(c)

$$\vec{a} = \ddot{\vec{r}} = -24e^{-2t}\hat{n}_r + 12e^{-2t}3\cos(\pi t)\hat{n}_\varphi + 12e^{-2t}3\cos(\pi t)\hat{n}_\varphi + 6(1-e^{-2t})[-3\pi\sin(\pi t)\hat{n}_\varphi + -9\cos(\pi t)\cos(\pi t)\hat{n}_r]$$

$$= [-24e^{-2t} - 54(1-e^{-2t})\cos^2(\pi t)]\hat{n}_r +$$

$$[36e^{-2t}\cos(\pi t) + 36e^{-2t}\cos(\pi t) - 18(1-e^{-2t})\pi\sin(\pi t)]\hat{n}_\varphi$$

$\vec{a} = -49.94\hat{n}_r - 9.74\hat{n}_\varphi \text{ cm/s}^2$ $|\vec{a}| = 50.88 \text{ cm/s}^2$



the angle between $\vec{v}$ and the horizontal line is $\arctan 84.43$ upwards

(d) the acceleration of the rod is equal to the acceleration of B in transverse component.

so $\vec{a}_{\text{relative}} = -49.9471 \text{ m/s}^2$ upwards.